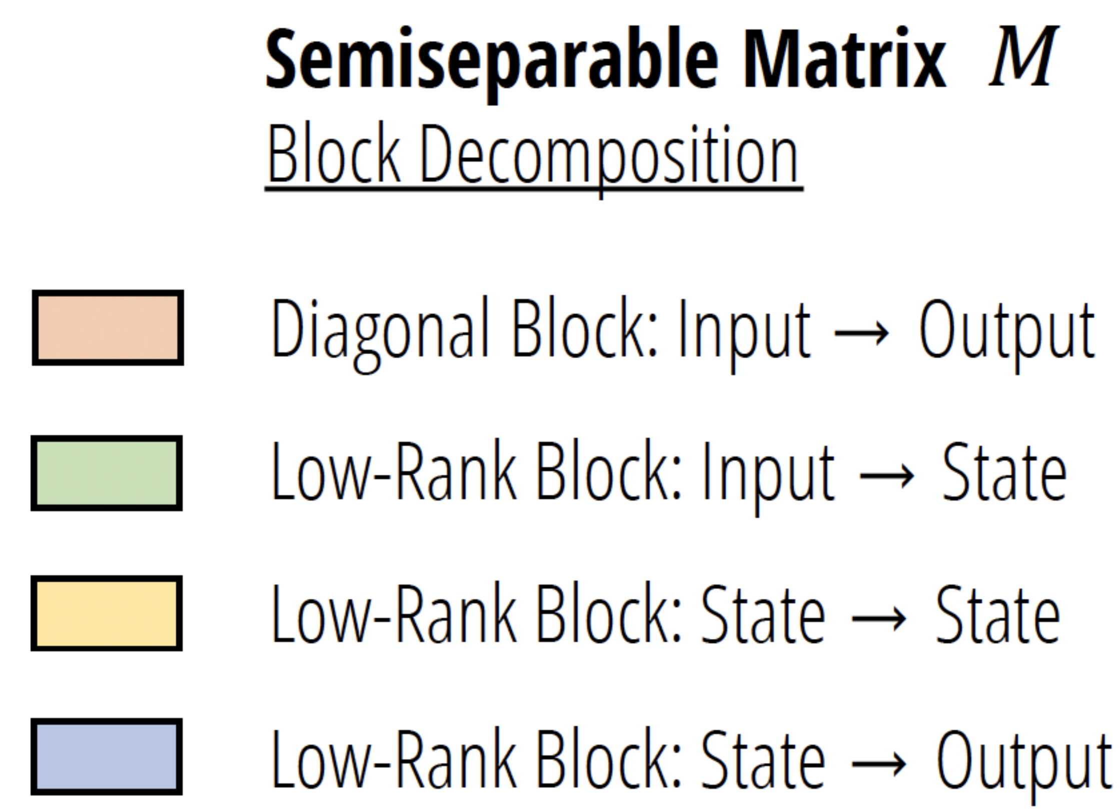
$$
M = \left[\begin{array}{ccc|ccc|ccc}
C_0 A_{0:0} B_0 & & & & & & & & \\
C_1 A_{1:0} B_0 & C_1 A_{1:1} B_1 & & & & & & & \\
C_2 A_{2:0} B_0 & C_2 A_{2:1} B_1 & C_2 A_{2:2} B_2 & & & & & & \\
\hline
C_3 A_{3:0} B_0 & C_3 A_{3:1} B_1 & C_3 A_{3:2} B_2 & C_3 A_{3:3} B_3 & & & & & \\
C_4 A_{4:0} B_0 & C_4 A_{4:1} B_1 & C_4 A_{4:2} B_2 & C_4 A_{4:3} B_3 & C_4 A_{4:4} B_4 & & & & \\
C_5 A_{5:0} B_0 & C_5 A_{5:1} B_1 & C_5 A_{5:2} B_2 & C_5 A_{5:3} B_3 & C_5 A_{5:4} B_4 & C_5 A_{5:5} B_5 & & & \\
\hline
C_6 A_{6:0} B_0 & C_6 A_{6:1} B_1 & C_6 A_{6:2} B_2 & C_6 A_{6:3} B_3 & C_6 A_{6:4} B_4 & C_6 A_{6:5} B_5 & C_6 A_{6:6} B_6 & & \\
C_7 A_{7:0} B_0 & C_7 A_{7:1} B_1 & C_7 A_{7:2} B_2 & C_7 A_{7:3} B_3 & C_7 A_{7:4} B_4 & C_7 A_{7:5} B_5 & C_7 A_{7:6} B_6 & C_7 A_{7:7} B_7 & \\
C_8 A_{8:0} B_0 & C_8 A_{8:1} B_1 & C_8 A_{8:2} B_2 & C_8 A_{8:3} B_3 & C_8 A_{8:4} B_4 & C_8 A_{8:5} B_5 & C_8 A_{8:6} B_6 & C_8 A_{8:7} B_7 & C_8 A_{8:8} B_8
\end{array}\right]
$$

$$
= \left[\begin{array}{c|c|c}
\begin{matrix} C_0 A_{0:0} B_0 & & \\ C_1 A_{1:0} B_0 & C_1 A_{1:1} B_1 & \\ C_2 A_{2:0} B_0 & C_2 A_{2:1} B_1 & C_2 A_{2:2} B_2 \end{matrix} & & \\
\hline
\begin{bmatrix} C_3 A_{3:2} \\ C_4 A_{4:2} \\ C_5 A_{5:2} \end{bmatrix} A_{2:2} \begin{bmatrix} B_0^\top A_{2:0} \\ B_1^\top A_{2:1} \\ B_2^\top A_{2:2} \end{bmatrix}^\top & \begin{matrix} C_3 A_{3:3} B_3 & & \\ C_4 A_{4:3} B_3 & C_4 A_{4:4} B_4 & \\ C_5 A_{5:3} B_3 & C_5 A_{5:4} B_4 & C_5 A_{5:5} B_5 \end{matrix} & \\
\hline
\begin{bmatrix} C_6 A_{6:5} \\ C_7 A_{7:5} \\ C_8 A_{8:5} \end{bmatrix} A_{5:2} \begin{bmatrix} B_0^\top A_{2:0} \\ B_1^\top A_{2:1} \\ B_2^\top A_{2:2} \end{bmatrix}^\top & \begin{bmatrix} C_6 A_{6:5} \\ C_7 A_{7:5} \\ C_8 A_{8:5} \end{bmatrix} A_{5:5} \begin{bmatrix} B_3^\top A_{5:3} \\ B_4^\top A_{5:4} \\ B_5^\top A_{5:5} \end{bmatrix}^\top & \begin{matrix} C_6 A_{6:6} B_6 & & \\ C_7 A_{7:6} B_6 & C_7 A_{7:7} B_7 & \\ C_8 A_{8:6} B_6 & C_8 A_{8:7} B_7 & C_8 A_{8:8} B_8 \end{matrix}
\end{array}\right]
$$



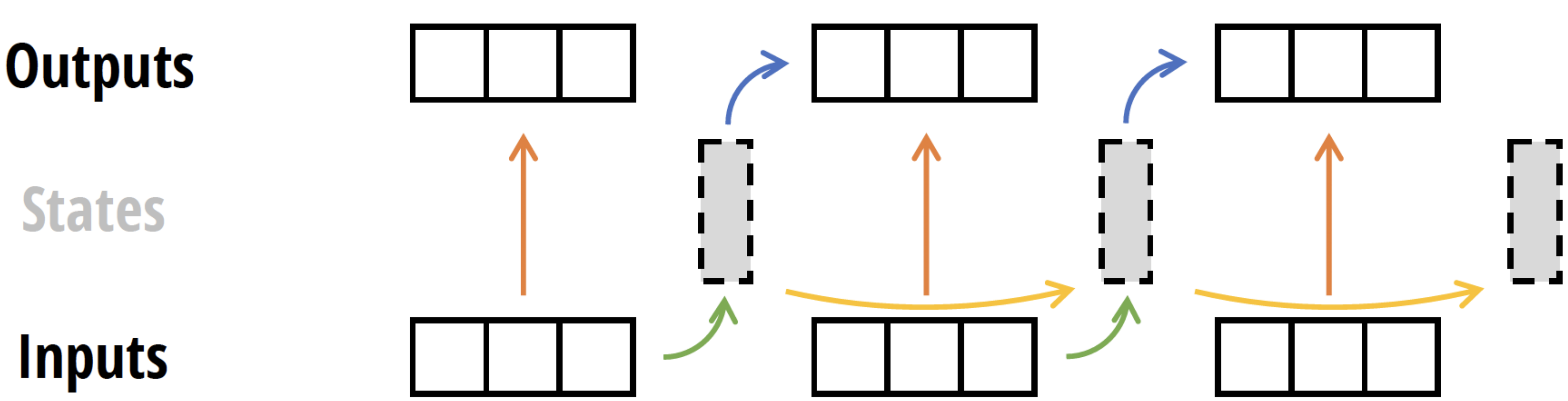


Figure 5: (**SSD Algorithm**.) By using the matrix transformation viewpoint of state space models to write them as semiseparable matrices (Section 3), we develop a more hardware-efficient computation of the SSD model through a block-decomposition matrix multiplication algorithm. The matrix multiplication also has an interpretation as a state space model, where blocks represent chunking the input and output sequence. Diagonal blocks represent intra-chunk computations and the off-diagonal blocks represent inter-chunk computations, factored through the SSM's hidden state.